Proving it works

The comparisons, the scenarios, the metrics, and what to do if negotiation loses.

> **AFTER THIS LESSON**
> You can explain what HarvestNet is compared against, what's measured, and why losing is an acceptable result.

Four methods, same inputs

- **First-come-first-served:** the first food bank to claim gets the whole batch. This is the headline comparison.
- **Nearest food bank.**
- **Rules-based negotiators:** the same process and calculators, with no AI.
- **LLM negotiators:** HarvestNet as designed.

Every method gets the same inputs at the same times, with the same rules and resources.

Five scenarios

A normal afternoon, more produce than storage, more households than produce, a cooler failure, and a condition doubt or no agreement.

What's measured

- **Produce wasted:** pounds unclaimed or spoiled. Proposed as a headline result.
- **Worst-served fair-share coverage:** how well the most-behind food bank's households did. Proposed as the second headline.
- **Households actually reached,** counted by confirmed pickups.
- Pickup-window match, time to commitment, and number-check failures.
- **Invalid commitments, which must be zero.**

If negotiation loses

Say so. The judges score "learning and evidence," and an honest result beats a tuned one. A useful finding might be: "Rules-based negotiation matched the LLM on waste. The LLM's value was reading messy input and explaining decisions."

Cost

A batch costs an estimated $0.50–1.00 in Claude calls, so the $100 credit covers roughly 100–200 batches. Rules-based runs and replays cost nothing, so use those for most testing. These are estimates, not measurements.

Source: docs/MVP_DESIGN.md, sections 14 and 16, and ADR 0005.

Read next: docs/MVP_DESIGN.md, section 16.

Previous: Lesson 9: No invented numbers · **Next:** Lesson 11: Architecture and stack

Quick lookup: HarvestNet at a glance.

Anything unclear? Ask me. I'm your teacher for this, and I can re-explain any part in different words.

Source of truth: docs/MVP_DESIGN.md, CONTEXT.md, and docs/adr/. If this lesson and those disagree, they win.